Problem 2: Number of Provinces (medium)

Problem Statement

Imagine a country with several cities. Some cities have direct roads connecting them, while others might be connected through a sequence of intermediary cities. Using a matrix representation, if `matrix[i][j]` holds the value `1`, it indicates that city `i` is directly linked to city `j` and vice versa. If it holds `0`, then there's no direct link between them.

Determine the number of separate city clusters (or provinces).

A **province** is defined as a collection of cities that can be reached from each other either directly or through other cities in the same province.

Examples

1. - **Input:**

     ```
     [[1,1,0],[1,1,0],[0,0,1]]
     ```

   - **Expected Output:**

     `2`

   - **Justification:**

     There are two provinces: cities 0 and 1 form one province, and city 2 forms its own province.

2. - **Input:**

     ```
     [[1,0,0,1],[0,1,1,0],[0,1,1,0],[1,0,0,1]]
     ```

   - **Expected Output:**

     `2`

   - **Justification:**

     There are two provinces: cities 0 and 3 are interconnected forming one province, and cities 1 and 2 form another.

3. - **Input:**

```
[[1,0,0],[0,1,0],[0,0,1]]
```

- **Expected Output:**

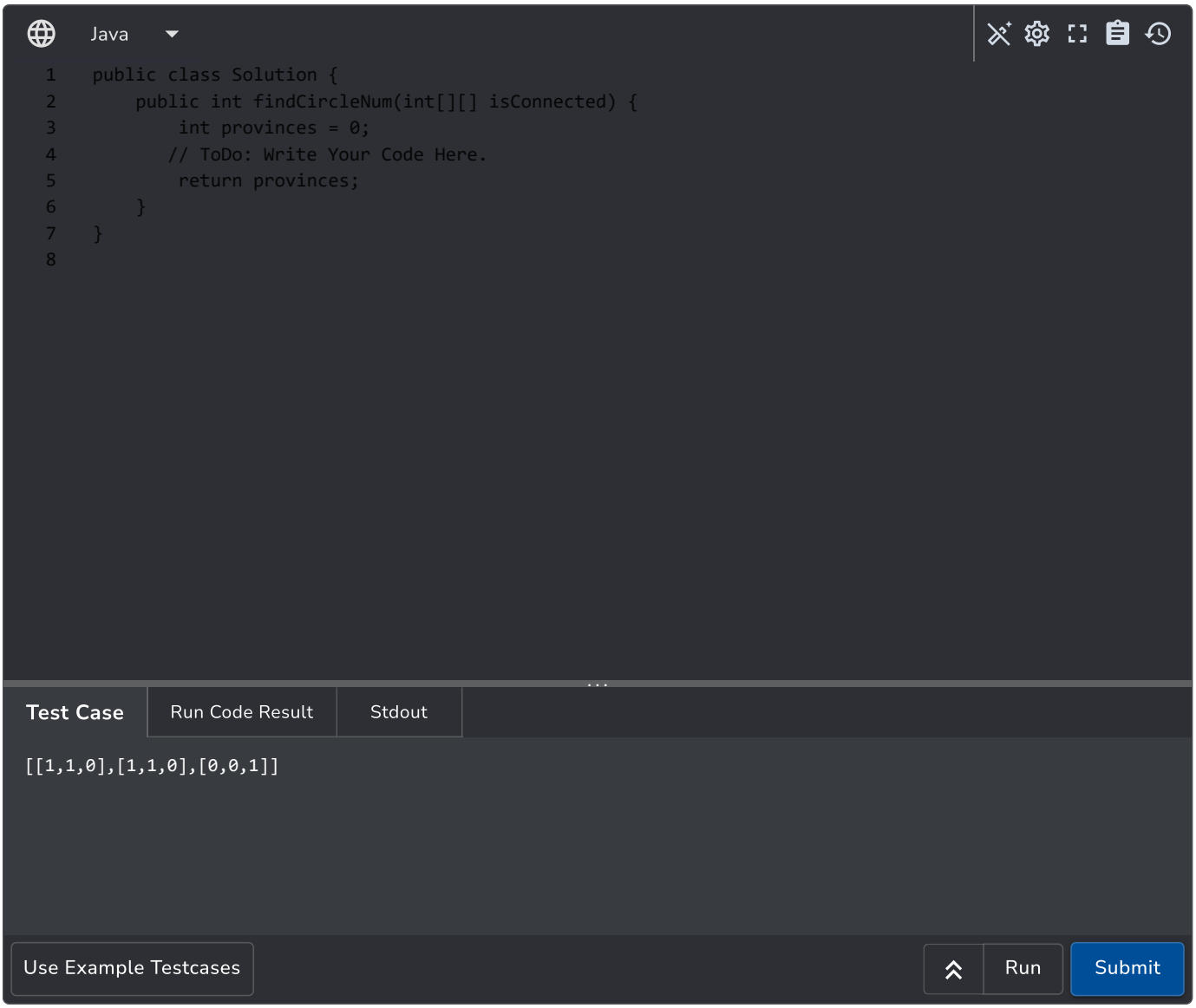
  `3`

- **Justification:**

  Each city stands alone and is not connected to any other city. Thus, we have three provinces.

Try it yourself

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